Introduction

Tree

Data Structures and its Applications (UE25CS52A)

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Quiz : Tree

1. The number of edges in a tree having n nodes

a) Depends whether the tree is a binary tree or not

b) Is always n - 1

c) Is 2n as each node has 2 pointers

d) Can vary from 1 to n

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Quiz : Tree

2. The longest path from the root to the leaf in any binary tree of n nodes has the length

a) n-1

b) log(n, 2)

c) n

d) n/2

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Quiz : Tree

3. The max number of leaves in a binary tree of n nodes is

a) n-1

b) log(n, 2)

c) Floor(n/2) + 1

d) Floor((n +1)/2)

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Quiz : Tree

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a) n-1

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c) Floor(n/2) + 1

d) Floor((n +1)/2)

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Quiz : Tree

4. In a binary search tree (key less than the root on the left), the node with the least key

a) Will be a leaf

b) Will not have a left sub tree

c) Will not have a right sub tree

d) Can have both left and right sub trees

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Quiz : Tree

4. In a binary search tree (key less than the root on the left), the node with the least key

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Quiz : Tree

5. The number of comparisons while searching a binary search tree of n nodes in the worst case will be

a) n

b) n/2

c) Log(n, 2)

d) 1

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Quiz : Tree

5. The number of comparisons while searching a binary search tree of n nodes in the worst case will be

a) n

b) n/2

c) Log(n, 2)

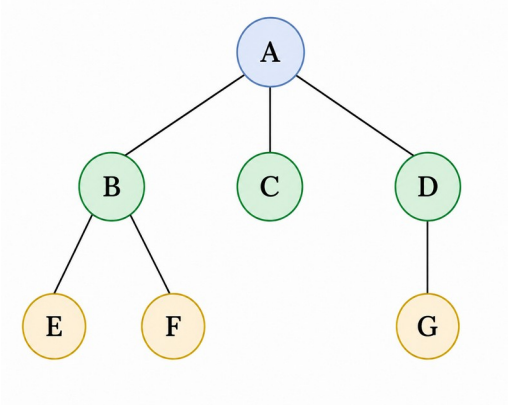
d) 1

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General Tree

Definition and Properties

- A **tree**
  - is a non-linear data structure used to represent hierarchical relationships
  - organizes elements in a parent–child structure.

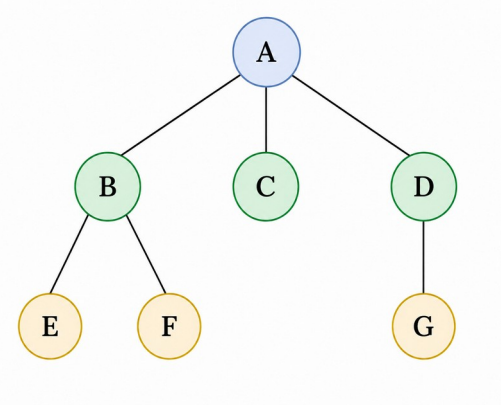


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General Tree

Terminology from genealogy:

- Parent
- Child
- Siblings
- Ancestor
- Descendant
- Uncle

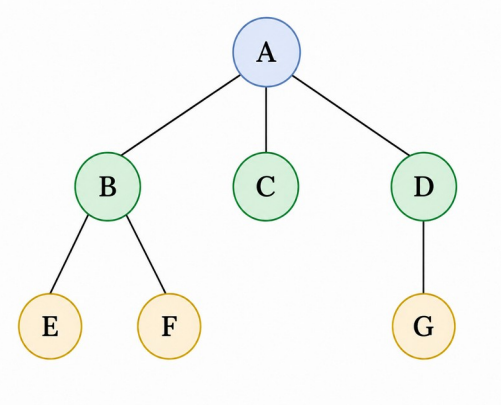


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General Tree

Terminology from botany:

- Root
- Nodes
- Leaves
- Internal nodes
- External nodes
- Branch

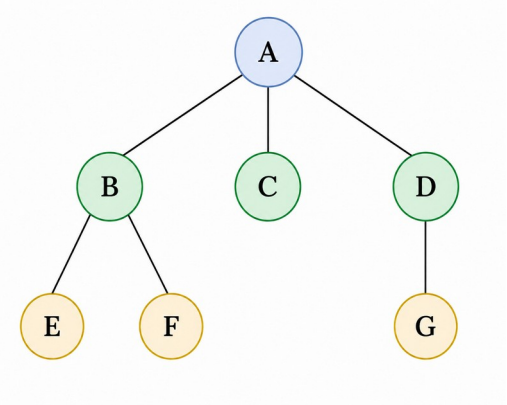


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General Tree

Definition

- A Tree
  - Is either empty
  - Has a root followed by number of trees
    - Called subtrees
- Some more terms
  - Path
  - Level
  - height

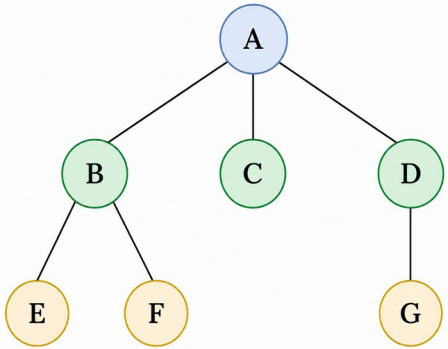


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General Tree

Important properties of trees:

- Acyclic : Can not start from a node, traverse and come back to the same node
- Connected. Any two nodes are connected by a path which may go through the root
- A tree of n nodes has n – 1 edges – neither more not less.
- Add an edge, we will have a cycle. Remove an edge, we will have more than one tree.

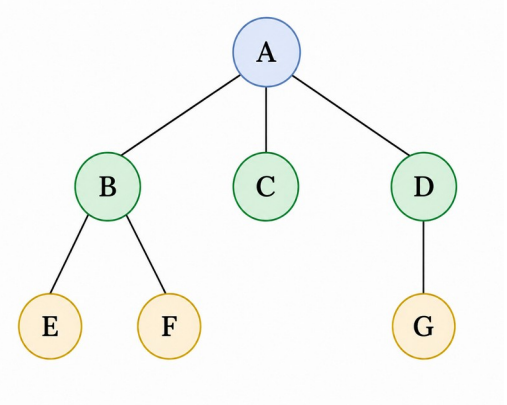


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General Tree

A possible ADT for a general tree :

- Create a tree
- Destroy the tree
- Check empty
- Check full
- Add a node
- Remove a node
- Traverse a tree
- Find root
- Find children
- Check whether leaf
- Find height of the tree
- Find the path from the root to a given node



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General Tree

Applications of the tree

- File System Structure:
- Search Trees:
- Databases, indexing and B-trees:
- Expression Evaluation:
- Compiler design:
- Document representation
- Heap Tree and priority queue:
- Process structure:
- Organization structure
- Decision Trees in AI
- ...

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General Tree

Applications of the tree

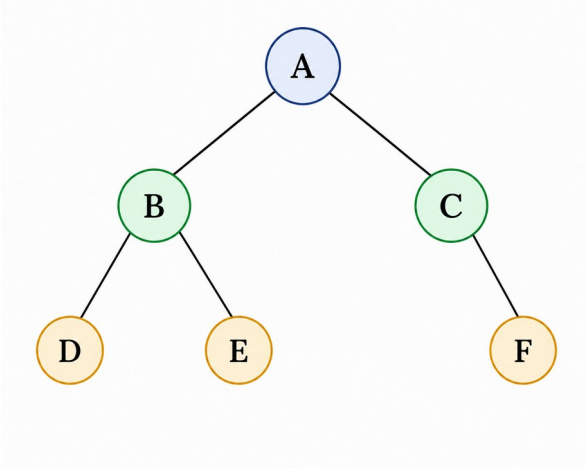
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Binary Tree

Definitiion

- A **binary tree** is a tree in which every node has **at most two children**.
- The two children are distinguished as the **left child** and **right child**.
- A binary tree is either:
  - An **empty tree**, or
  - A root node together with two disjoint binary trees called its:
    - **left subtree**
    - **right subtree**



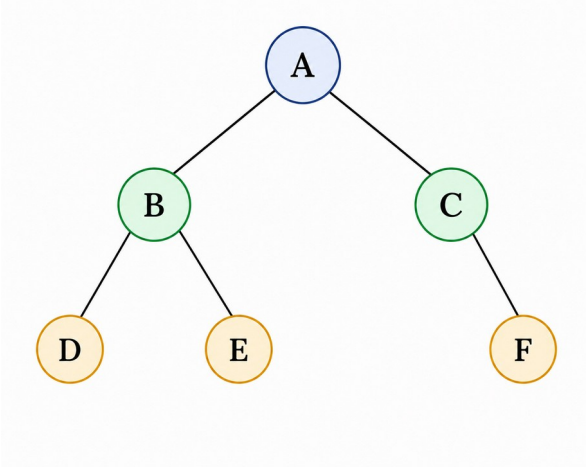
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Binary Tree

Inorder traversal of the binary tree

- If tree is not empty
  - Visit the left subtree
  - Visit the root
  - Visit the right subtree

Inorder traversal of this tree:
D B E A C F



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Binary Tree

Binary representation of any general tree

- Root of general tree becomes root of the binary tree
- Child of a node in general tree appears to the left of the node in the binary tree
- Sibling of a node in general tree will appear to the right of the node in the binary tree

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Binary Search Tree

- Every node has a key
- Nodes are inserted based on the key
- If the key of a node is less than that of the root, it will be inserted in the left subtree
- Otherwise, it will be inserted in the right subtree

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Binary Tree

Binary Search Tree : simple and small interface

- Create a tree
- Destroy the tree
- Insert the key
- Display the tree in inorder

- Find the number of nodes
- Find the number of leaves

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Binary Search Tree : structure of the functions

- Recursive on nodes and not on tree
- Model for recursive routine :
  - Empty tree
  - Tree with left and right subtrees

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THANK YOU

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